

Experiment1

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1 Part 1

Report: Plot.

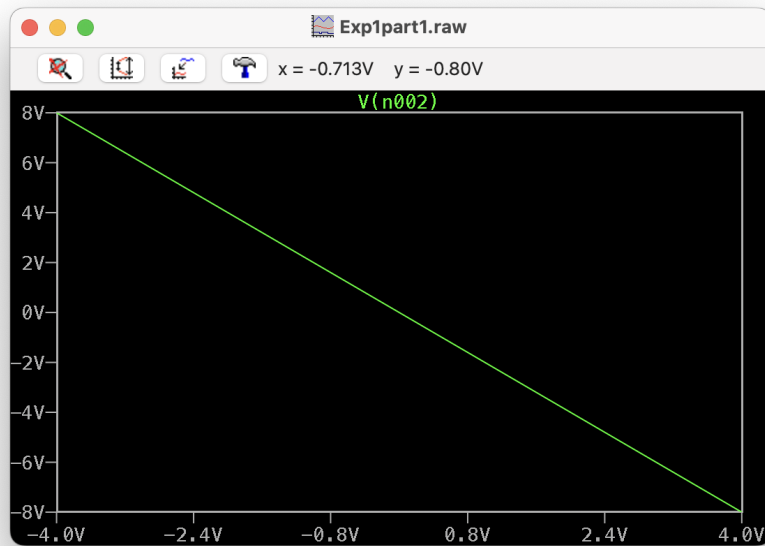


Figure 1: Gain = -2 verified

2 Part 2

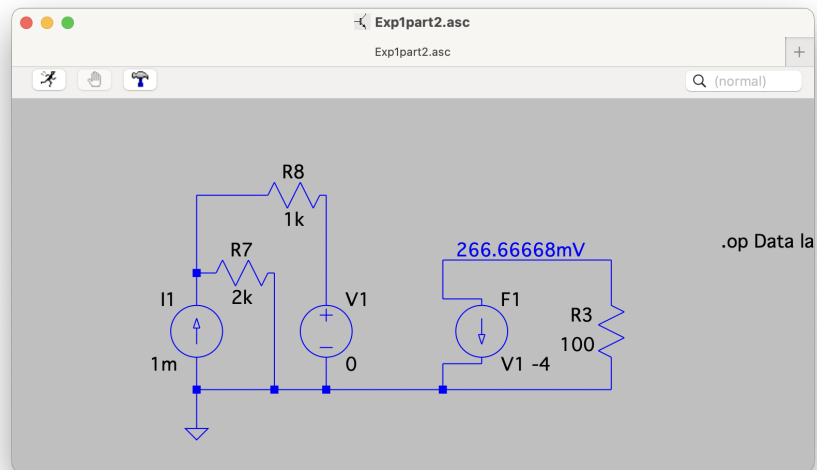


Figure 2: Schematic with updated output reading. Notice that I swap the original position of R_7 and R_8 to measure the current through R_8 equivalently.

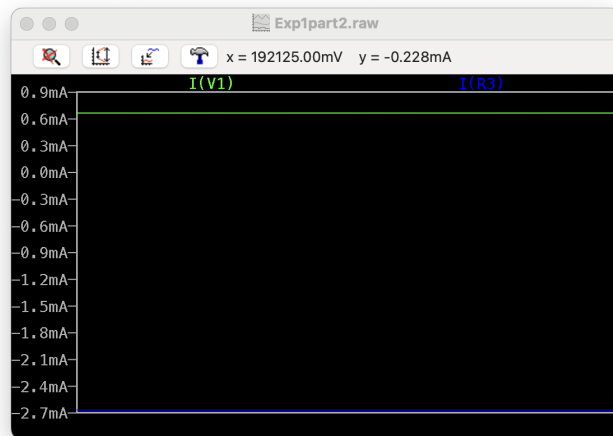


Figure 3: The result is verified that the output current of CCCS is -4 times of the current through R_8

3 Part 3

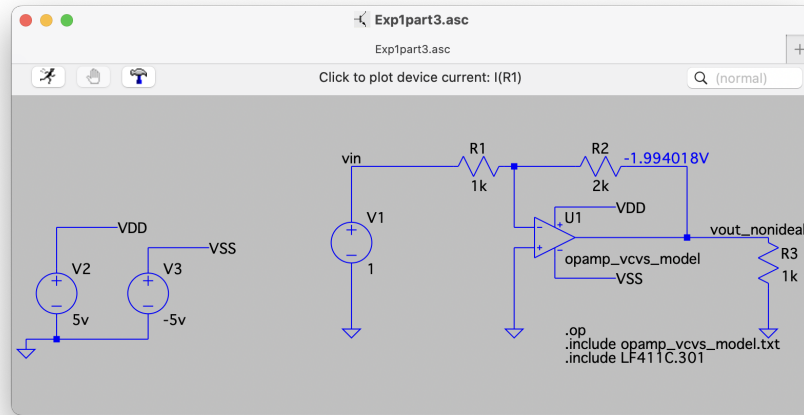


Figure 4: Original gain = 1000. $V_{out_nonideal} = -1.994018$ V

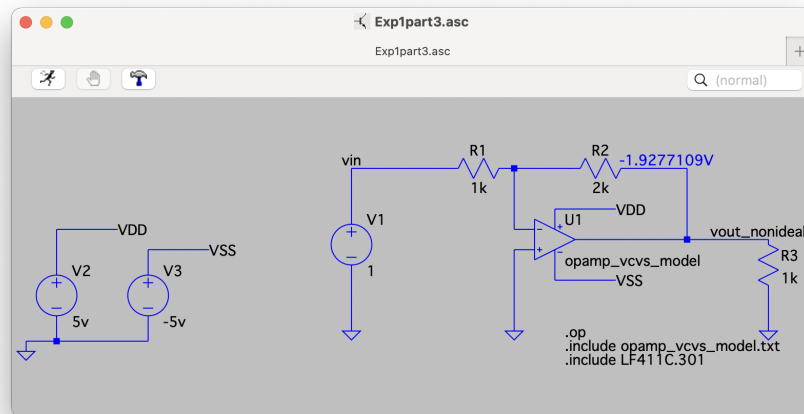


Figure 5: Gain = 80. $V_{out_nonideal} = -1.9277109$ V

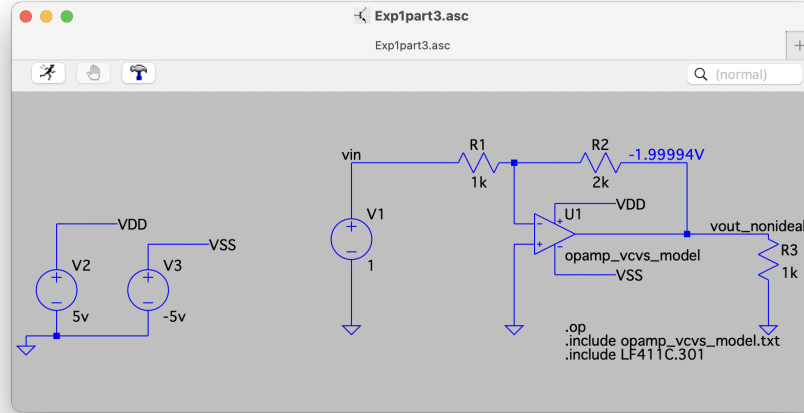


Figure 6: Gain = $1e5$. $V_{out_{nonideal}} = -1.99994$ V

$$A_{cl(ideal)} = -\frac{R_f}{R_{in}} = -\frac{2k\Omega}{1k\Omega} = -2$$

With an input voltage of +1 V , the ideal output should be:

$$V_{out (ideal)} = -2 \times 1 \text{ V} = -2 \text{ V}$$

High Gain ($1e5$):

- Output: - 1.99994 V
- Error: Only 0.06 mV from ideal (-2V)
- Comment: With high open-loop gain, the circuit closely approximates the ideal inverting amplifier equation

Medium Gain (1000):

- Output: -1.994018 V
- Error: 5.98 mV from ideal
- Comment: Moderate gain deviation from ideal behavior

Low Gain (80):

- Output: -1.9277109V
- Error: 72.29 mV from ideal
- Comment: Low open-loop gain significantly degrades circuit performance

The actual closed-loop gain approaches the ideal value when the open-loop gain is much larger than the desired closed loop gain.

4 Part 4

The saturation effect has been achieved. -5V to 5V

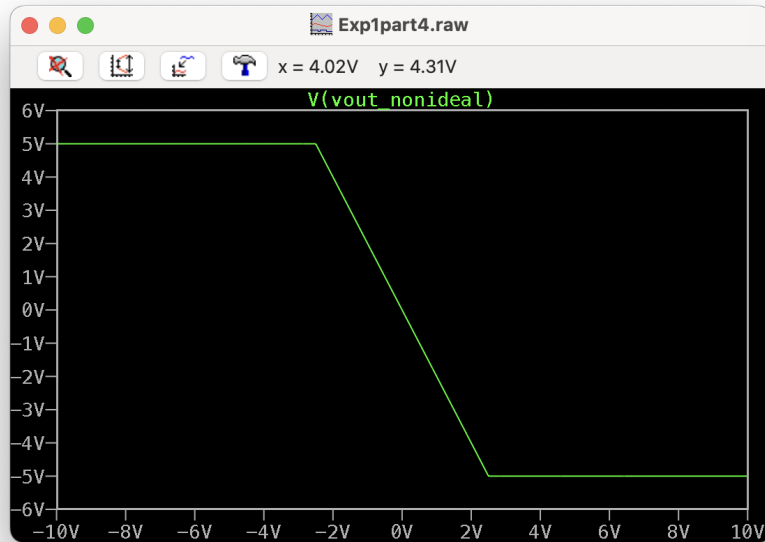


Figure 7: The saturation effect has been achieved. -5V to 5V

5 Part 5

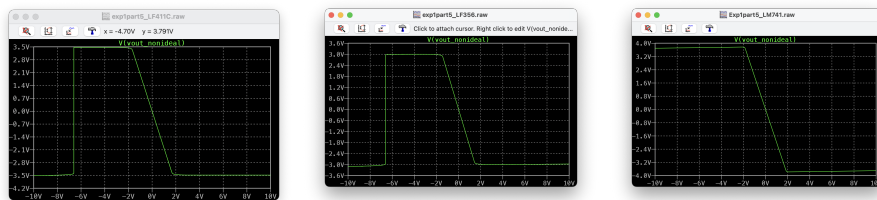


Figure 8: Supply Voltage $\pm 5V$ for 3 op amps.

LM741 gives the most ideal output and has the widest output range before saturation when the supply voltage is $\pm 5V$.

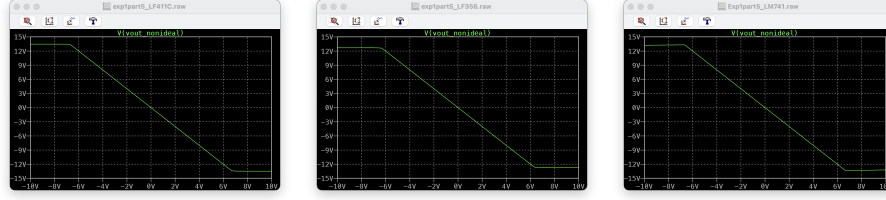


Figure 9: Supply Voltage $\pm 15V$ for 3 op amps.

LM741/LF411C gives the most ideal output and has the widest output range before saturation when the supply voltage is $\pm 15V$.

6 Part 6

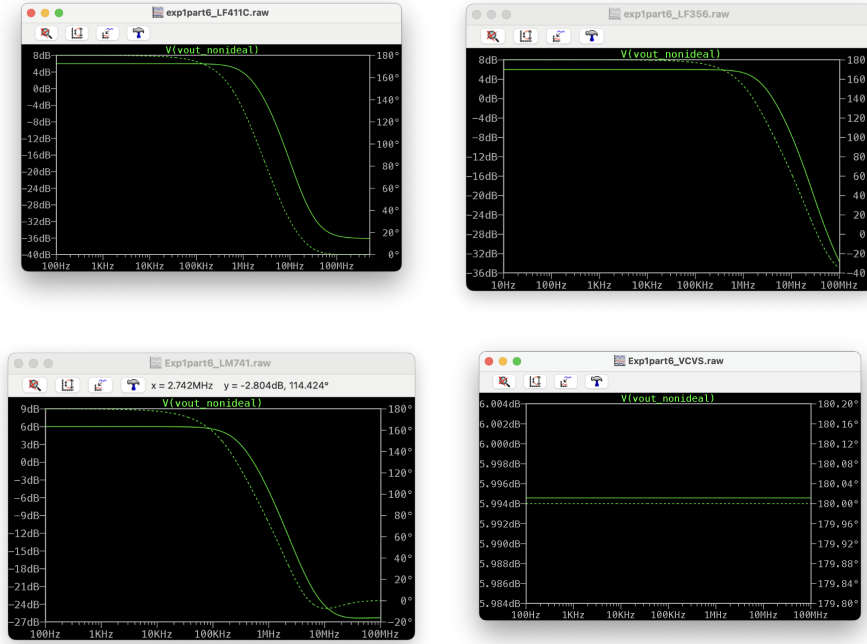


Figure 10: Top left LF411C. Top right LF356. Bottom left LM741. Bottom right VCVS

As the result shows, our VCVS does not have frequency limitations as the gain and phase are straight lines in the Bode plot while other commercial models have decaying gain and phase shift respectively.

LF 356 has yields the widest frequency response.

7 Part 7

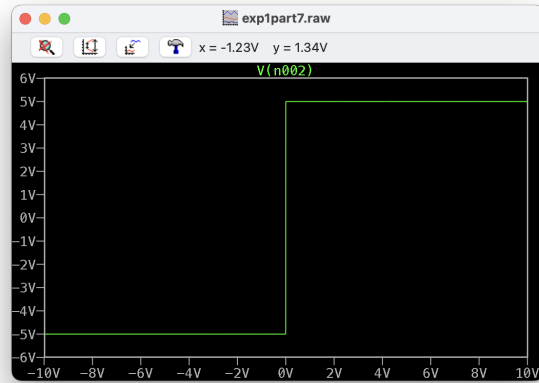


Figure 11: Plot of part 7

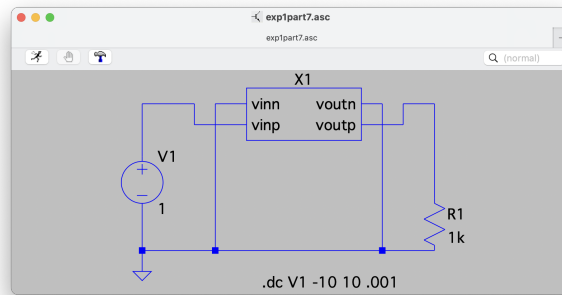


Figure 12: Schematic

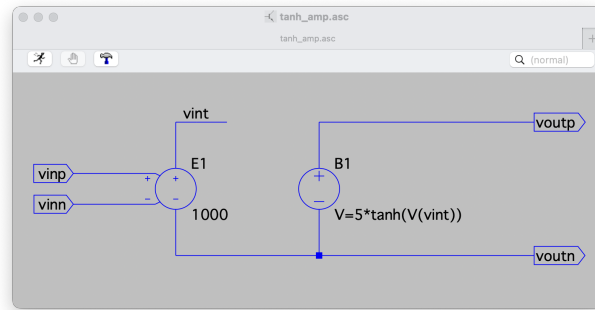


Figure 13: tanh amp Schematic